GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. A finite observer must acquire the information that it compresses and must pay for the decision to stop acquiring it. We study positive causal experiments with a finite affine closure of executable future tests, allowing a fixed unknown model and a single model-independent observer. Under a bound on expected acquisition, we characterize the attainable terminal predictive ensembles by a joint report-routing separation condition. No uniform transience or contraction is required. Excess invariant occupation can be removed without changing the terminal law or increasing any nonnegative resource cost; consequently the optimal hard-state risk is attained. The same result gives a finite-model obstruction to every unattainable robust risk level. For experiments whose current task response is recoverable from the current report, a physically prepared reference law yields a sharper geometric characterization. Its mass-constrained quantization profile gives an all-observer converse and a one-live-state implementation. For independent reference reports both checkpoint and autonomous optima are exact; for uniformly comparable controlled reports the bounds match. On an acquired s-regular support the excess risk is of order $(NM)^{-2/s}$, with every live and terminal state charged. Correlated classical kernels and noncommuting quantum instruments verify the same theorem. Singular product measures, changing-rank mixtures and attained calibration and simulation errors are treated explicitly.

1. Introduction

The geometry relevant to a prediction problem is the geometry that an observer actually acquires. For a causal experiment this geometry depends on preparation, intervention and stopping, as well as on the physical kernel. A cover of a formally reachable set is not an acquisition procedure. Conversely, a procedure that realizes a favorable conditional distribution may consume many unsuccessful observations before producing its report. The central question is to determine the predictive information that can be acquired and retained by an executable observer, with these costs included.

We take the experiment, not a manifold or an information matrix, as primitive. A preparation and positive report instruments determine future-test evaluations and hence a predictive quotient. A finite observer has a fixed program: it selects an action from its current live label, reads the next report, and either changes live label or enters a terminal label. The latter decision is part of the program. The audit is performed only afterwards. The program cannot read the true model, its own unrecorded past, or an external clock. Persistent labels, program constants, temporary arithmetic, calibration and physical calls are different resources.

Two difficulties arise when an acquisition budget is a mean constraint $\mathbb{E}\tau \le N$. First, an algebraic balance equation need not describe the occupation of the prepared run. It can include an invariant component not generated by the preparation. Second, optimizing the stopping rule changes the acquired law itself. A fixed-time posterior quantization formula therefore does not solve the stopped problem. These are different obstructions: the first concerns causal implementation, while the second concerns the geometry on which that implementation spends its budget.

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Theorem 3.2 treats both at their appropriate levels. On a finite affine family of executable tests, a joint report-routing inequality characterizes terminal predictive ensembles under the mean budget. Its witnesses specify one homogeneous finite-state program. Positivity permits removal of every finite invariant occupation component without changing the terminal ensemble. The budget then provides compactness, even when other programs never terminate. This proves attainment, and a model-uniform version retains the same program across an arbitrary family of dominated experiments. The associated strict risk obstructions already occur on finite subfamilies.

A sharp geometric consequence requires additional information about what a report reveals. Suppose the current task response is recoverable from the current report and the conditional one-call laws are comparable to a reference law ν that is itself physically prepared. Stopping with mean budget N can increase the density of an acquired terminal law by at most a factor proportional to N. The optimal distortion under this density constraint is a mass-constrained quantization profile $\mathcal{Q}_m^\nu(N)$. A rejection rule realizes the matching upper with one live state and m terminal states. The independent-preparation frontier of the retained adapted-stopping article [7], reproduced here as a limiting case, gives the exact identities

$$\mathcal{R}_{\mathrm{cp}}(N,m) - B = \mathcal{Q}_m^\nu(N), \qquad \mathcal{R}_{\mathrm{aut}}(N,M) - B = \mathcal{Q}_{M-1}^\nu(N). \tag{1.1}$$

For correlated controlled reports the profile remains a two-sided comparison. If the acquired reference support is s-regular, it yields $(NM)^{-2/s}$ rather than a rate determined by the ambient parameter dimension. There is no requirement that s be an integer. Atoms and rank changes are governed by the profile, not by an artificially assigned common dimension.

The distinction between the two levels is essential. The occupation theorem applies to informative normalized filters with a finite affine closure; the sharp profile theorem does not silently assume that such a filter is report-recoverable. Its classical realization is a genuinely correlated controlled observation experiment, and its quantum realization is a noncommuting instrument with continuously many outcomes. Neither realization is used in the proof of the general theorem. Known-kernel filtering, stationary observation, scheduled horizons and free stopping are separate tasks; a rate in one is not asserted in another without a causal comparison.

Relation to prior work. Predictive states and minimal future-test representations have an established history; see Shalizi and Crutchfield [8] and the experiment-comparison treatment of Fritz, Gonda, Perrone and Rischel [3]. Finite belief compression is also classical, including the value-directed approach of Poupart and Boutilier [5]. These works motivate, but do not supply, the hard-state mean-acquisition theorem proved here.

Occupation equations and constrained control are classical tools. In particular, Dufour and Prieto-Rumeau [2] characterize the invariant part of phantom occupation solutions. We do not claim that decomposition as new. Here it is used for a different purpose: a finite positive predictive representation, a report-only homogeneous controller, and terminal risk under an upper bound on all acquired occupation. One may retain a compact algebraic relaxation and remove its invariant part rather than assume uniform absorption. The proof must still enforce one routing program, including its stopping decision, across all models. A recent preprint of Sokolov [9] gives a Markovization of stopping intensities while preserving observed and surviving measures. A policy of the form $\lambda(t, X_t)$ has access to a state and time; it is not, without further work, a controller with a prescribed finite number of persistent labels.

The functional $\mathcal{Q}_m^\nu$ is normalized trimmed quantization, not a new static quantization objective. Cuesta-Albertos, Gordaliza and Matrán [1] established trimmed k-means, and Vérine, Pydi, Negrevergne and Chevaleyre [11] study budgeted rejection sampling for distributional objectives. The exact independent-preparation value and its one-live-state realization are inherited from [7], whose theorem already allows a nonconstant conditional Bayes risk and singular preparation. The extension here is the joint model-independent routing theorem under a direct mean budget, together with a common-program geometric comparison for correlated controlled model families at the same

hard-state budget. The small-ball and covering proof is included for every real acquisition budget, so no asymptotic quantization theorem is being invoked outside its hypotheses.

The retained adapted-stopping article [7] also proves a distinct theorem for informative Gaussian and noncommuting weak-effect instruments: at each fixed positive acquisition price it matches the optimal full-history sensing-and-stopping value with a finite-state value. That proof uses calibrated occupation and cost absorption, not report recoverability. We retain it intact rather than identify a Lagrange price with a constrained mean budget without a duality theorem. Its error-composition and calibration mechanisms are inherited tools. Appendix C likewise reproduces the earlier Bayesian information decomposition [6]. The new realizations below extend the constrained comparison to correlated classical and unknown noncommuting report-recoverable instruments; they do not establish the same sharp constrained curve for the retained Gaussian filter.

Bayes-risk quantization and Bregman geometry are prior constructions [10]. Kim's finite-memory study [4] compares fixed-policy costs under one closed-loop execution. Here the converse ranges over all declared adaptive acquisition and stopping policies, while the implementing rule is one common finite-state program. These different quantifiers, and not the mere use of a belief approximation, delimit the contribution.

Scope. The paper proves a class theorem, not a finite-dimensional reduction of every causal experiment. Finite affine closure, common report domination and the audit interface are explicit in Section 2. Report recoverability and uniform comparison are additional hypotheses only for the sharp profile. There is no assumption of a preassigned Fisher matrix, no use of total variation to infer a large-deviation principle, and no promotion of finite matrices to unbounded-operator identities. An actual mean budget is not an almost-sure deadline. Computational evaluation of the exact occupation optimum is a further problem; the existence theorem is not a polynomial-time synthesis claim.

2. Prepared experiments and finite observers

2.1. **The raw experiment and its predictive representation.** Let the hidden state, report and history spaces be standard Borel. A preparation and a finite action set $\mathcal{A}$ specify normalized positive causal kernels

$$K_a^\theta(dx', dy \mid x), \qquad \theta \in \Theta.$$

The model θ is fixed during a run and is not an input to the observer. The singleton case is included. Every chosen action incurs one acquisition. The observer may stop after any acquired report, but not before the first. After stopping, a specified protected audit is executed: it cannot be used to decide whether to continue. Fresh randomization is independent of the hidden experiment conditional on the visible history.

We impose the following finite predictive representation, to be verified from the raw kernels. Let E be a finite-dimensional real vector space, $C \subset E$ a closed, pointed, generating cone, and $e \in E^*$ strictly positive on $C \setminus \{0\}$. Its base

$$D = \{x \in C : e(x) = 1\}$$

is compact. A preparation is $x_0^\theta \in D$. For each action, Y_a is standard Borel and λ_a is a finite positive measure. There are measurable positive linear maps

$$L_a^\theta(y) : C \longrightarrow C, \qquad \int \|L_a^\theta(y)\| \lambda_a(dy) < \infty, \qquad \int e L_a^\theta(y) x \, \lambda_a(dy) = e(x). \tag{2.1}$$

The measures λ_a are common to all models, but the integrability constants need not be uniform in θ. Starting from x, the report has density $eL_a^\theta(y)x$ and the updated predictive state is

$$F_a^\theta(y, x) = \frac{L_a^\theta(y)x}{eL_a^\theta(y)x} \tag{2.2}$$

on positive-evidence reports. If the denominator is zero, positivity implies $L_a^\theta(y)x = 0$; no positive-probability posterior is assigned there.

This representation means that the selected executable future tests, their report pullbacks and the audit moments belong to a finite affine closure. Its linear evaluations must separate the predictive quotient under consideration. The closure is a hypothesis, not a consequence of finite physical dimension alone. Appendix A gives a measurable construction and minimality statement, including the distinction between the actual quotient and the compact convex base used to certify positive mixtures.

For the audit, let $Z \in \mathbb{R}^k$ be bounded. There are linear maps $T_\theta : E \to \mathbb{R}^k$ and $c_\theta \in E^*$ such that, at predictive state $x \in D$,

$$\mathbb{E}_\theta[Z \mid x] = T_\theta x, \qquad \mathbb{E}_\theta[\|Z\|^2 \mid x] = c_\theta(x). \tag{2.3}$$

The decoder lies in a common compact convex set $H \subset \mathbb{R}^k$ containing all possible conditional means. The loss is $\|Z - r\|^2$, uniformly bounded by a declared constant L. All models have this same report, action, audit and loss interface. The unnormalized terminal cost is the continuous function

$$G_\theta(d, r) = c_\theta(d) - 2\langle r, T_\theta d\rangle + e(d)\|r\|^2, \qquad d \in C, \quad r \in H. \tag{2.4}$$

For a known single model, minimizing in r gives the continuous perspective

$$\overline{g}(d) = c(d) - \frac{\|Td\|^2}{e(d)}, \quad e(d) > 0, \qquad \overline{g}(0) = 0. \tag{2.5}$$

The continuity at zero follows from compactness of D. In a model family the minimization in r cannot be performed separately for each model.

2.2. **What the machine may retain.** A homogeneous autonomous observer has a nonempty live set S and a nonempty terminal set J, disjoint, with

$$|S| + |J| \le M. \tag{2.6}$$

Its initial live label is s_0. A live label s selects an action according to p_{sa}. Having read y, the observer routes to $i \in S \sqcup J$ with Borel probabilities

$$\gamma_i(s, a, y) \ge 0, \qquad \sum_{i \in S \sqcup J} \gamma_i(s, a, y) = 1. \tag{2.7}$$

An arrival in J halts and supplies decoder $r_j \in H$. The stop/continue decision and the destination are one joint kernel. All p, γ, r are independent of θ and of the time index. Private random bits are fresh; a retained random seed would require an additional state. The report and selected action can be processed within the current transition, but no uncharged value survives it. In particular, a terminal label is not a live label together with a free external halting bit.

Let $\tau \ge 1$ be the number of acquisitions before the first terminal label. A program is admissible at budget N if

$$\sup_{\theta \in \Theta} \mathbb{E}_\theta \tau \le N < \infty. \tag{2.8}$$

This implies almost-sure termination for each model. No statement is made about programs outside this admissible class. Define

$$\mathcal{R}^\Theta_{\mathrm{aut}}(N, M) = \inf_{(p,\gamma,r) \text{ satisfying } (2.6),(2.8)} \sup_{\theta \in \Theta} \mathbb{E}_\theta \|Z - r_{J_\tau}\|^2.$$

For $N \ge 1, M \ge 2$ the class is nonempty: acquire once and halt at a fixed decoder. For $N < 1$ or $M < 2$ it is empty under this interface. Its infimum is then $+\infty$, a convention about admissibility, not a divergent statistical loss.

A checkpoint with m terminal labels may use the entire visible history for acquisition and stopping, subject to the same mean constraint, and then encode the stopped history into one of m

labels. Only this final encoding is constrained by m. This stronger architecture has no claimed free implementation by an m-state autonomous observer.

2.3. **Acquired occupation and terminal ensembles.** For one fixed program let $X_t^\theta \in D$ denote the exact predictive state after t reports. Define

$$z_s^{\theta,\mathrm{act}} = \mathbb{E}_\theta \sum_{t=0}^{\tau-1} \mathbf{1}_{\{S_t=s\}} X_t^\theta, \qquad\qquad d_j^{\theta,\mathrm{act}} = \mathbb{E}_\theta[\mathbf{1}_{\{J_\tau=j\}} X_\tau^\theta]. \tag{2.9}$$

These are vector integrals in the positive cone. Their masses are actual visit counts and stopping probabilities:

$$\sum_s e(z_s^{\theta,\mathrm{act}}) = \mathbb{E}_\theta \tau, \qquad \sum_j e(d_j^{\theta,\mathrm{act}}) = 1.$$

The vectors in (2.9) are proof objects, not registers in the observer. When solving algebraic balance equations below, a candidate z_s^θ must not be called an actual occupation until its preparation-generated part has been identified.

3. A budgeted causal acquired-geometry theorem

We first define the finite-dimensional feasible object in the theorem. Fix live and terminal sets S, J and $s_0 \in S$. Put

$$a_s^\theta = \mathbf{1}_{\{s=s_0\}} x_0^\theta.$$

A candidate consists of common action probabilities p_{sa}, common decoders $r_j \in H$, and, for each θ, vectors $z_s^\theta, d_j^\theta \in C$ with

$$z_s^\theta - a_s^\theta \in C, \qquad \sum_s e(z_s^\theta) \le N, \qquad \sum_j e(d_j^\theta) = 1. \tag{3.1}$$

For every finite $F \subset \Theta$ and every collection $u_s^\theta, v_j^\theta \in E^*$, require

$$\begin{aligned} &\sum_{\theta\in F}\left[\sum_{s\in S}\langle u_s^\theta, z_s^\theta - a_s^\theta\rangle + \sum_{j\in J}\langle v_j^\theta, d_j^\theta\rangle\right] \\ &\quad\le \sum_{s\in S,\, a\in\mathcal{A}} p_{sa} \int \max\left\{\max_{i\in S}\sum_{\theta\in F}\langle u_i^\theta, L_a^\theta(y) z_s^\theta\rangle, \max_{j\in J}\sum_{\theta\in F}\langle v_j^\theta, L_a^\theta(y) z_s^\theta\rangle\right\}\lambda_a(dy). \end{aligned} \tag{3.2}$$

The summation over models precedes the maximum. The maximum chooses one next label, including the decision to stop, and cannot be chosen separately for different models. Define

$$\Phi^\Theta(N,M) = \min \sup_{\theta\in\Theta}\sum_{j\in J} G_\theta(d_j^\theta, r_j), \tag{3.3}$$

where the minimum is over the finitely many cardinalities $|S|, |J| \ge 1$, $|S| + |J| \le M$, and over candidates satisfying (3.1)–(3.2). The minimum and its operational meaning are part of the following result.

For the geometric part, let $K \subset \mathbb{R}^k$ be compact and ν a probability measure with support K. Write

$$\mathcal{Q}_m^\nu(t) = \min_{c_1,\dots,c_m \in \operatorname{conv} K} \min_{\substack{0\le h\le t \ (\nu\text{-a.e.}) \\ \int h\, d\nu = 1}} \int_K \min_{1\le j\le m} \|x - c_j\|^2 h(x)\nu(dx), \qquad t \ge 1. \tag{3.4}$$

Both minima are attained, as proved below. The factor t is a constraint on acquired probability mass, not a radius or a dimension.

Definition 3.1 (Report-recoverable task with comparable acquisition)**.** A task is report-recoverable if, whenever action a produces report y, its current conditional audit mean is a specified Borel function $x = f_a(y) \in K$, independent of the incoming history and of θ, and its conditional audit variance is a common constant B. Suppose ν is the actual one-call law under a specified admissible reference preparation and action. There are constants $0 < a_* \le 1 \le b_* < \infty$ such that, for every model and every positive-probability visible history, every legal next action has task-response law at most $b_*\nu$. One fixed available action a_0, common to all models, has conditional task-response law at least $a_*\nu$ at every such history. These are measure inequalities, not pointwise assertions on null histories.

The lower inequality is needed only for the implementing action. A raw-kernel bound on a closed invariant base is one way, used in the realizations, to verify the required history quantifiers. The actual terminal law remains policy-dependent.

Theorem 3.2 (Budgeted causal acquired-geometry transfer)**.** *The following statements hold.*

(i) General implementation and robust attainment. *Under* (2.1)–(2.4)*, for every real $N \ge 1$ and integer $M \ge 2$,*

$$\mathcal{R}^{\Theta}_{\mathrm{aut}}(N, M) = \Phi^{\Theta}(N, M). \tag{3.5}$$

A common Borel program attains this value. No uniform transience, positive stopping floor, mixing assumption, or finite bound on τ is imposed. In addition,

$$\mathcal{R}^{\Theta}_{\mathrm{aut}}(N, M) = \sup_{\substack{F \subset \Theta \\ 0 < |F| < \infty}} \mathcal{R}^{F}_{\mathrm{aut}}(N, M). \tag{3.6}$$

Thus every risk level strictly below the full-family optimum is already impossible on some finite model subfamily, with the same N, M and one common program.

(ii) Actual-law geometric bounds. *For a report-recoverable task as in Definition 3.1, all fully adaptive checkpoint policies and all admissible autonomous programs satisfy*

$$\mathcal{Q}^{\nu}_{m}(b_*N) \le \mathcal{R}^{\Theta}_{\mathrm{cp}}(N, m) - B, \tag{3.7}$$

$$\mathcal{Q}^{\nu}_{M-1}(b_*N) \le \mathcal{R}^{\Theta}_{\mathrm{aut}}(N, M) - B. \tag{3.8}$$

If $a_*N \ge 1$, one common autonomous program gives

$$\mathcal{R}^{\Theta}_{\mathrm{aut}}(N, M) - B \le \frac{b_*}{a_*}\mathcal{Q}^{\nu}_{M-1}(a_*N). \tag{3.9}$$

It has exactly one live state and at most $M-1$ terminal states, and its stopping time has a conditional geometric tail with mean at most N. The analogous checkpoint upper uses m terminal states in place of $M-1$.

If every conditional response law equals ν, independently of all previous reports and all actions, then for all $N \ge 1$,

$$\mathcal{R}^{\Theta}_{\mathrm{cp}}(N, m) - B = \mathcal{Q}^{\nu}_{m}(N), \qquad \mathcal{R}^{\Theta}_{\mathrm{aut}}(N, M) - B = \mathcal{Q}^{\nu}_{M-1}(N). \tag{3.10}$$

The realizing rule in this case has $\mathbb{E}\tau = N$, including noninteger N.

(iii) Matching resource law on acquired regular supports. *Suppose in addition that, for some $s > 0$, $r_0 > 0$ and $0 < c_0 \le C_0 < \infty$,*

$$c_0 r^s \le \nu(B(x, r)) \le C_0 r^s \quad (x \in K,\ 0 < r \le r_0). \tag{3.11}$$

Then, simultaneously for all $N \ge 1$, $m \ge 1$ and $M \ge 2$,

$$\mathcal{R}^{\Theta}_{\mathrm{cp}}(N, m) - B \asymp (Nm)^{-2/s}, \tag{3.12}$$

$$\mathcal{R}^{\Theta}_{\mathrm{aut}}(N, M) - B \asymp \big(N(M-1)\big)^{-2/s}. \tag{3.13}$$

The constants depend only on $a_*, b_*, c_0, C_0, r_0, s$ and the diameter of K, not on N, m, M, the number of models, or the chosen adaptive policy.

Part (i) is the causal implementation theorem. Part (ii) uses a stronger report interface to turn implementation into a sharp acquired-geometry statement. Definition 3.1 by itself does not require a finite affine closure; when both sets of hypotheses hold, the profile bounds are bounds on the exact value (3.3). A general normalized hidden-state filter need not satisfy recoverability. Part (iii) is not an assignment of dimension to arbitrary reachable states: its measure is an explicitly obtained reference law and its lower bound covers the actual stopped law of every competing policy.

Remark 3.3 (What an occupation certificate certifies). A candidate z^θ in (3.1) can contain invariant mass. The theorem certifies its terminal ensemble d^θ, with an actual occupation no larger than z^θ. It does not identify every candidate with the acquired occupation. This distinction is precisely what permits compactness without uniform absorption. Equality constraints on expected acquisition, rewards for spending resources, and negative running costs cannot be substituted in this argument.

Remark 3.4 (Fixed-time and stopping architectures). The $M-1$ in (3.10) pays for one live state. It is not the $M-n$ deduction for an internally counted, deterministic deadline. The implementing program uses no clock at all. It may run for arbitrarily many acquisitions, each charged in $\mathbb{E}\tau$. A comparison with a pathwise cap, a latency quantile, or a prescribed audit time requires a different resource constraint.

4. Common routing and removal of invariant occupation

We prove Theorem 3.2(i) without using any realization or any metric geometry.

Lemma 4.1 (A joint routing separation criterion)**.** *Fix common p and vectors $z_s^\theta \in C$. There exist common Borel routing probabilities* (2.7) *such that, for all θ,*

$$z_i^\theta - a_i^\theta = \sum_{s,a} p_{sa} \int \gamma_i(s,a,y) L_a^\theta(y) z_s^\theta \, \lambda_a(dy) \qquad (i \in S), \tag{4.1}$$

$$d_j^\theta = \sum_{s,a} p_{sa} \int \gamma_j(s,a,y) L_a^\theta(y) z_s^\theta \, \lambda_a(dy) \qquad (j \in J) \tag{4.2}$$

if and only if (3.2) *holds for every finite F and every displayed dual vector.*

Proof. For each pair (s,a) the simplex-valued kernels form a weak-star compact convex subset of $L^\infty(\lambda_a)^{S \sqcup J}$. Positivity and the sum-one constraint are weak-star closed: test them against nonnegative L^1 functions. The product over finitely many (s,a) is compact and convex. Integration against $L_a^\theta(y) z_s^\theta$ is weak-star continuous by (2.1). For a finite F, the vector of the right sides of (4.1)–(4.2) therefore has a compact convex image.

Its support function is the right side of (3.2). Indeed, after summing the paired contributions over $\theta \in F$, one maximizes a linear function over a probability simplex, for each (s,a,y). A maximizing label can be selected by the smallest-index rule among finitely many measurable functions. This gives a Borel maximizer and the stated integral of a maximum. Finite-dimensional separation proves that the target vector belongs to the image exactly when all support inequalities hold.

For an arbitrary Θ, impose the balance equations first on each finite F. Their solution sets in the same weak-star compact routing space are closed and have the finite intersection property: the union of finitely many finite model sets is finite. A common routing kernel exists by compactness. On standard Borel report spaces its equivalence classes admit Borel versions. Changing representatives on a common λ_a-null set and assigning a fixed destination there makes (2.7) pointwise valid without changing any model law. Necessity is integration of the pointwise simplex bound. □

For such a kernel define positive linear maps on ensembles,

$$(P_\theta z)_i = \sum_{s,a} p_{sa} \int \gamma_i(s,a,y) L_a^\theta(y) z_s \, \lambda_a(dy), \quad (Q_\theta z)_j = \sum_{s,a} p_{sa} \int \gamma_j(s,a,y) L_a^\theta(y) z_s \, \lambda_a(dy).$$

Write $\boldsymbol{e}(z) = \sum_s e(z_s)$, with the analogous convention on terminal ensembles. Normalization gives

$$\boldsymbol{e}(P_\theta z) + \boldsymbol{e}(Q_\theta z) = \boldsymbol{e}(z) \quad \text{for every positive live ensemble } z. \tag{4.3}$$

Lemma 4.2 (Budget-preserving removal)**.** *Let P, Q be positive ensemble maps satisfying* (4.3)*. If a finite positive ensemble solves $z = a + Pz$, then*

$$h = \lim_{n\to\infty} P^n z \in C^S, \qquad Ph = h, \qquad Qh = 0, \qquad z^{\mathrm{act}} = \sum_{n=0}^{\infty} P^n a = z - h. \tag{4.4}$$

The common observer started at preparation a has actual continuing ensemble z^{act}, terminal ensemble Qz, and mean acquisition $\boldsymbol{e}(z^{\mathrm{act}}) \le \boldsymbol{e}(z)$. Every positive linear running-cost functional is no larger on the actual ensemble.

Proof. Since $Pz = z - a \le_C z$, the sequence $P^n z$ decreases in the cone order. Its scalar mass decreases to a finite limit. Compactness of the base implies an estimate $\|w\| \le C_D \boldsymbol{e}(w)$ for positive ensembles. Thus, for $n \ge m$,

$$\|P^m z - P^n z\| \le C_D(\boldsymbol{e}(P^m z) - \boldsymbol{e}(P^n z)),$$

so the sequence converges in the finite-dimensional space. Its limit h is positive, and continuity of P gives $Ph = h$. Applying (4.3) to h gives $\boldsymbol{e}(Qh) = 0$. Strict positivity of the mass functional on the cone implies $Qh = 0$, component by component.

Iteration of $z = a + Pz$ yields

$$z = \sum_{n=0}^{t-1} P^n a + P^t z.$$

Taking the limit proves (4.4). Starting the physical experiment at a, induction with the raw instruments shows that $P^n a$ is exactly the unnormalized predictive ensemble of surviving live labels after n acquisitions. This statement uses only linearity before normalization and the fact that the same report-routing kernel is executed at every step. Tonelli's theorem, applied first to masses and then to the bounded base coordinates, gives the actual occupation and $\mathbb{E}\tau = \boldsymbol{e}(z - h) < \infty$. Hence $\tau < \infty$ almost surely. Its terminal ensemble is $Q\sum_n P^n a = Q(z - h) = Qz$. A positive linear cost decreases when the positive component h is removed. □

Example 4.3 (Why the candidate is not the acquired law)**.** Take the scalar cone $\mathbb{R}_+$, live states $0, 1$, terminal state 2, preparation $a = (1, 0)$. State 0 always halts after one call; state 1 always returns to itself. Then

$$P = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad Q = (1, 0), \qquad z = (1, h), \quad d = 1$$

solves the balance equations for every $h \ge 0$. The prepared run never visits state 1. Its mean acquisition is 1, not $1 + h$. The extra component changes neither the terminal law nor an upper budget constraint. This is the elementary phantom-measure phenomenon in occupation theory [2]; its explicit role here is to delimit the interpretation of a predictive certificate.

Proof of Theorem 3.2(i). For an admissible observer, linear conditional integration and Tonelli give (4.1)–(4.2) with the actual ensembles (2.9). The mean budget gives (3.1); Lemma 4.1 gives (3.2). Conditional expectation at the protected audit gives risk $\sum_j G_\theta(d_j^\theta, r_j)$. Therefore $\Phi^\Theta \le \mathcal{R}^\Theta_{\mathrm{aut}}$.

Conversely, a feasible candidate has one common routing witness by Lemma 4.1. For each θ, Lemma 4.2 identifies the actual ensemble generated by its preparation. The same kernel, not a model-specific choice of kernel, is used for every θ. Its actual expected acquisition is at most N,

and its terminal ensemble is exactly d^θ. Its loss is therefore the candidate loss. This proves the reverse inequality, including at the level of infima.

It remains to show attainment. For fixed cardinalities, action rows lie in compact simplices and decoders in H^J. For each model, the positive ensemble sets in (3.1) are compact, because their masses are bounded and D is compact. Take the product of these compact sets over Θ. Each inequality (3.2) depends on finitely many models and is continuous in p, z, d. To verify continuity of its integral, bound the difference of maxima by the largest difference of the finitely many linear terms, and use the integrability in (2.1). The feasible set is therefore compact. It is nonempty by the one-call stopping program. The objective, a supremum of continuous model losses, is lower semicontinuous; it attains its minimum. There are only finitely many cardinality choices. The implementing kernel supplied above attains the resulting global minimum.

For (3.6), the inequality from right to left follows by restriction. Suppose that r is strictly smaller than the full-family optimum and no finite subfamily has optimum greater than r. Work in the finite disjoint union of the compact candidate spaces, adding the closed constraints $\sum_j G_\theta(d_j^\theta, r_j) \le r$. Any finite collection of support inequalities and risk constraints refers to finitely many models. A minimizing candidate for their union satisfies the collection. Coordinates for other models may be filled arbitrarily in their nonempty budget sets. Thus these closed sets have the finite intersection property and yield a full-family feasible candidate of risk at most r, a contradiction. This also proves the strict finite-model obstruction. There is no interchange of inf and sup, and no bound on the size of the obstructing finite set is asserted. □

Corollary 4.4 (Additional nonnegative costs)**.** *In Theorem 3.2(i), finitely many further constraints of the form*

$$\sum_{s,a} p_{sa}\, \kappa^\theta_{\ell,sa}(z_s^\theta) \le B_\ell,$$

where each $\kappa^\theta_{\ell,sa}$ is a fixed nonnegative linear functional representing an actual conditional expected primitive cost, can be added. The unit acquisition bound is retained. The value equality and finite-model statement remain valid, and the value is attained whenever the constrained admissible class is nonempty.

Proof. The additional inequalities are closed. After extracting a common routing witness, removal of its invariant occupation can only decrease each displayed cost. The compactness and finite-intersection arguments are unchanged. □

Remark 4.5 (Necessity versus a finite computation)*.* The support inequalities are an exact realizability criterion, not a finite numerical test in general. Infeasibility for a fixed candidate and finite model family has a strict finite-dimensional separating vector. The global finite-model conclusion is stronger but still non-effective: it supplies neither a uniformly bounded witness size nor an algorithm for locating it. These distinctions prevent a compactness proof from being reported as a finite-bit synthesis theorem.

5. Actual stopped mass and matching geometry

The independent-report specialization below is the constrained-frontier mechanism already proved in [7]; the comparison with correlated, controlled, model-indexed conditional rows is the extension. We give the argument in full because the history quantifiers and stopping-cost accounting are essential. The reference measure in this section is a one-call acquired law, obtained from the reference preparation in Definition 3.1. It is not an assumption that any stopped policy has this law.

Lemma 5.1 (Compact mass-constrained quantization)**.** *For every compact $K \subset \mathbb{R}^k$, probability ν supported on K, $m \ge 1$ and real $t \ge 1$, the minimum in* (3.4) *exists. For fixed centers put $d_C(x) = \min_j \|x - c_j\|$. A minimizing density may be chosen in the form*

$$h(x) = t\mathbf{1}_{\{d_C(x)^2 < \lambda\}} + t\xi \mathbf{1}_{\{d_C(x)^2 = \lambda\}}, \qquad 0 \le \xi \le 1, \tag{5.1}$$

with $\int h\,d\nu = 1$. *The case* $t = 1$ *permits* $h = 1$.

Proof. Centers may be projected onto the compact convex set $\operatorname{conv} K$, which cannot increase any distance to K. The set of densities $0 \le h \le t$, $\int h = 1$, is weak-star compact in $L^\infty(\nu)$. If centers converge, their squared nearest-center distance functions converge uniformly on K. The objective is therefore jointly continuous for this convergence and weak-star convergence of h. Compactness gives a minimum.

For fixed centers choose λ such that

$$t\nu(d_C^2 < \lambda) \le 1 \le t\nu(d_C^2 \le \lambda)$$

and choose ξ to fill the possible boundary mass. Against any other admissible density $\widetilde{h}$, one has

$$\int (d_C^2 - \lambda)(\widetilde{h} - h)\,d\nu \ge 0,$$

because on the sublevel set $h = t$ and on the strict superlevel set $h = 0$. The equal total masses remove the λ term. This proves optimality, including atoms and positive boundary mass. At a joint minimum the centers can also be replaced by the conditional means of their selected Voronoi cells, with any fixed Borel rule for ties; the squared-error projection identity proves that this does not increase the objective. □

Lemma 5.2 (Acquisition mass bound)**.** *For any model-independent adaptive history policy and randomized stopping rule with* $\mathbb{E}_\theta\tau \le N$, *under the upper comparison in Definition 3.1,*

$$\operatorname{Law}_\theta(X_\tau) \le b_* N \nu. \tag{5.2}$$

More precisely, the unnormalized law of all acquired responses before and at stopping satisfies

$$\mathbb{E}_\theta \sum_{t=1}^{\tau} \mathbf{1}_{\{X_t \in A\}} \le b_* \nu(A) \mathbb{E}_\theta \tau \quad (A \text{ Borel}). \tag{5.3}$$

Proof. The event $\{\tau \ge t\}$ is determined before acquisition t, including all preceding private coins. Conditional on this information and the chosen action, the law of X_t is at most $b_*\nu$. Summing

$$\mathbb{E}_\theta[\mathbf{1}_{\{\tau \ge t\}} \mathbf{1}_{\{X_t \in A\}}] \le b_* \nu(A) \mathbb{P}_\theta(\tau \ge t)$$

over $t \ge 1$ proves (5.3) by Tonelli. The terminal indicator is no larger than the acquired-count sum. Its total mass is one, because finite mean implies almost-sure stopping, which proves (5.2). □

Lemma 5.3 (One-live-state realization)**.** *Let* $a_* N \ge 1$. *Given centers* $C = \{c_1, \dots, c_m\}$ *and* $0 \le h \le a_* N$ *with* $\int h\,d\nu = 1$, *execute the fixed action* a_0 *and, after reading* $x = f_{a_0}(y)$, *accept with probability*

$$A(x) = \frac{h(x)}{a_* N}. \tag{5.4}$$

On acceptance, enter the nearest-center terminal label; otherwise return to the single live label. For every model this program satisfies

$$\mathbb{E}_\theta\tau \le N, \qquad \mathbb{E}_\theta d_C(X_\tau)^2 \le \frac{b_*}{a_*} \int d_C^2 h\,d\nu. \tag{5.5}$$

If all conditional response laws equal ν, *the choice* $A = h/N$ *with* $0 \le h \le N$ *gives* $\mathbb{E}\tau = N$ *and* $\operatorname{Law}(X_\tau) = h\nu$ *exactly.*

Proof. At each surviving history the acceptance probability is at least

$$a_* \int \frac{h}{a_* N}\,d\nu = \frac{1}{N}.$$

Consequently $\mathbb{P}_\theta(\tau > n) \le (1 - 1/N)^n$ and $\mathbb{E}_\theta \tau \le N$. The conditional expected squared distortion contributed by acceptance at that call is at most

$$\frac{b_*}{a_* N} \int d_C^2 h \, d\nu.$$

Summing it over surviving histories gives the second inequality of (5.5). This proof does not assume that repeated reports are independent or that the conditional law after rejection equals the original law.

For independent reference rows, the probability of acceptance at each surviving call is exactly $1/N$, and its accepted sublaw is $h\nu/N$. The sum of the geometric series is N, proving both exact assertions. Equivalently, the conditional response kernel being ν at every history suffices; no additional hidden-state independence is used. □

Proof of Theorem 3.2(ii). The protected audit identity gives, for every legal stopped policy and decoder,

$$\mathbb{E}_\theta \|Z - r_J\|^2 = B + \mathbb{E}_\theta \|X_\tau - r_J\|^2. \tag{5.6}$$

For a checkpoint there are at most m decoder values. For an autonomous observer there are at most $M - 1$, since at least one state is live. Conditional randomization of the decoder cannot improve squared loss; its mean belongs to H. Projecting centers onto $\operatorname{conv} K$ can only help. Lemma 5.2 bounds the actual terminal density by $b_* N$, so pointwise nearest-center comparison proves (3.7)–(3.8) for every model and every randomized observer. It therefore proves the robust lower as well.

Apply Lemma 5.3 to a minimizer in Lemma 5.1, with $m = M - 1$. This is a common program, determined by ν, the comparison constants and its centers, not by θ. It proves (3.9). The same construction is available in the more permissive checkpoint architecture. If the conditional laws all equal ν, the terminal law is exactly the minimizing $h\nu$, and the upper equals the lower. This proves (3.10). The stop is executed before the audit; no future audit information enters the accepting set. □

Lemma 5.4 (Regular-support profile)**.** *Under* (3.11)*,*

$$c\,(tm)^{-2/s} \le \mathcal{Q}_m^\nu(t) \le C\,(tm)^{-2/s} \quad (t \ge 1,\ m \ge 1), \tag{5.7}$$

with constants depending only on the data specified in Theorem 3.2(iii).

Proof. For any center $c \in \operatorname{conv} K$, if $B(c, r)$ meets K, choose $x \in K \cap B(c, r)$. Then $K \cap B(c, r) \subset B(x, 2r)$. Hence, when $2r \le r_0$, any union of m such balls has ν-mass at most $mC_0(2r)^s$. An admissible density is at most t. Choose a fixed sufficiently small $c_1 > 0$ and set $r = c_1(tm)^{-1/s}$, so that this acquired mass is at most $1/2$ and $2r \le r_0$. At least half the selected probability lies at distance at least r, proving the lower bound.

For the upper we record the covering and packing steps. A maximal r-separated subset of K covers K by r-balls, and its disjoint $r/3$-balls have mass at least $c_0(r/3)^s$. Thus its cardinality is at most $3^s/(c_0 r^s)$. Enlarging constants to include the finitely small-cardinality regime gives a global cover by at most m balls of radius $C_1 m^{-1/s}$.

Conversely, choosing $\delta = c_2 m^{-1/s}$ with c_2 sufficiently small, fewer than m balls of radius δ cannot cover K, by the upper mass bound. A greedy construction therefore produces m centers separated by at least δ. For $t \ge t_0$, where t_0 depends only on the regularity constants, set

$$r = (c_0 m t)^{-1/s} \le \delta/3.$$

The m disjoint r-balls have union U of mass at least $mc_0 r^s = 1/t$. The density $h = \mathbf{1}_U/\nu(U)$ is admissible and its distance to these centers is at most r. This gives the required upper for $t \ge t_0$. For $1 \le t < t_0$, use $h = 1$ and the global m-center cover: $C_1^2 m^{-2/s} \le C_1^2 t_0^{2/s} (tm)^{-2/s}$. All changes for radii exceeding r_0 are absorbed by the fixed diameter constant, not by t or m. □

Proof of Theorem 3.2(iii). For $a_*N \geq 1$, insert Lemma 5.4 into the two profile bounds. For $1 \leq N < 1/a_*$, stop after one report of a_0 and use the global m-center cover. This has mean acquisition one and excess at most $Cm^{-2/s}$, which is at most $Ca_*^{-2/s}(Nm)^{-2/s}$ in this range. The lower profile estimate holds for all $N \geq 1$. The state deduction $m = M - 1$ is exact throughout. □

Proposition 5.5 (Atoms retain their physical mass)**.** *For any compact acquired reference law and $m \geq 1, t \geq 1$,*

$$\mathcal{Q}_m^{\nu}(t) = 0 \quad \Longleftrightarrow \quad \text{some set of at most } m \text{ atoms has total } \nu\text{-mass at least } 1/t. \tag{5.8}$$

In particular, an atom of mass $w > 0$ gives zero excess in the independent-reference experiment once $N \geq 1/w$, with one live and one terminal state.

Proof. At a minimum with zero distortion, the probability measure $h\nu$ is supported on the at most m centers. It can carry unit mass only if their ν-mass is at least $1/t$. Conversely, restrict and normalize ν to such a finite set; the resulting density is at most t and its distortion at those centers is zero. The causal assertion follows from the exact rejection implementation. All rejected observations are included in the mean $1/w$. □

6. Two raw-kernel realizations

The next results verify the preceding general theorem from physical kernels. Their finite affine representations are derived here and are not used as premises in Sections 4–5.

6.1. **Correlated controlled geometric observations.** Fix $d \geq 1$, let $K \subset [-1/2, 1/2]^d$ be compact, and let ν be a centered probability with support K. The physical state space is the whole box; preparation at the origin is allowed even when the origin is outside K. A finite set of controls has unknown, fixed coefficients $\theta_a \in [-\eta, \eta]$, where $0 \leq \eta < 1$. One raw acquisition from physical state x selects the new state y and reports it exactly, with kernel

$$K_a^{\theta}(dx', dy \mid x) = k_a^{\theta}(x, y)\, \delta_y(dx')\nu(dy), \qquad k_a^{\theta}(x, y) = 1 + \frac{4\theta_a}{d}\langle x, y\rangle. \tag{6.1}$$

Centering normalizes the kernel and $1 - \eta \leq k_a^{\theta} \leq 1 + \eta$. The next report is generally correlated with the previous physical state. Different control coefficients change its conditional law. No reset to ν is made after a rejection. Preparation at $x = 0$ produces ν in one actual call, uniformly in θ.

The protected audit has conditionally independent coordinates $Z_i \in \{-1, 0, 1\}$, with

$$\mathbb{P}(Z_i = \pm 1 \mid x) = \frac{x_i^2 + 1/2 \pm x_i}{2}, \qquad \mathbb{P}(Z_i = 0 \mid x) = \frac{1}{2} - x_i^2. \tag{6.2}$$

All these probabilities are positive on the box. Direct calculation gives $\mathbb{E}[Z_i \mid x] = x_i$ and $\mathrm{Var}(Z_i \mid x) = 1/2$.

Theorem 6.1 (Classical realization)**.** *The prepared kernels* (6.1)–(6.2) *satisfy the finite affine and model-independent hypotheses of Theorem 3.2(i), and Definition 3.1 with $a_* = 1 - \eta$, $b_* = 1 + \eta$ and $B = d/2$. If ν satisfies* (3.11)*, then the same theorem yields, for the entire controlled unknown-coefficient family,*

$$\mathcal{R}_{\mathrm{aut}}^{\Theta}(N, M) - \frac{d}{2} \asymp \big(N(M-1)\big)^{-2/s}. \tag{6.3}$$

For uniform measure on the box, $s = d$. Every mean-budget lower holds for arbitrary history-dependent sensing and randomized stopping, not only for the implementing fixed action.

Proof. Let $\phi(x) = (x^{\alpha})_{\alpha \in \{0,1,2\}^d}$, with the constant coordinate first, and let $D = \mathrm{conv}\, \phi([-1/2, 1/2]^d)$. Work on its linear span with cone generated by D and mass equal to the constant coordinate. The

base is compact; the cone is closed and pointed. With z_{e_i} denoting a first-moment coordinate, define the raw instrument

$$L_a^\theta(y)z = \left(z_0 + \frac{4\theta_a}{d}\sum_i y_i z_{e_i}\right)\phi(y). \tag{6.4}$$

It is positive on the cone because its scalar coefficient is the integral of the positive density (6.1) under a positive mixture of physical states. It is integrable and normalized. Its normalized output is $\phi(y)$, independently of the incoming predictive mixture. This verifies both finite affine update closure and report recoverability.

Every audit event has a probability polynomial of degree at most two in each coordinate, hence an affine evaluation on D. Conversely, the three univariate audit probabilities span $1, x_i, x_i^2$. Their products show that the finite audit-event family separates every coordinate of D. Future finite report-and-action words are obtained by positive compositions and integrations of (6.4). Thus this is a finite closure of executable tests, not an arbitrary feature matrix. The task map extracts the first moments, and its second-moment evaluation is

$$c(\phi(x)) = \frac{d}{2} + \sum_i x_i^2.$$

On the actually acquired output $\phi(y)$, the conditional Bayes risk is $d/2$.

The raw likelihood bound $1-\eta \le k \le 1+\eta$ survives conditioning on any history, since conditional mixtures preserve these measure inequalities. It holds for every adaptive choice of the next action. The first-call reference law is exactly ν, and every fixed action is suitable for the upper construction. For completeness, the report kernel is continuous in the physical input, with

$$\|K_a^\theta(x,\cdot) - K_a^\theta(x',\cdot)\|_{\mathrm{TV}} \le \frac{2\eta}{d}\int |\langle x-x', y\rangle|\,\nu(dy).$$

On the full convex base the analogous bound uses the difference of first moments. Same-report predictive updates have zero discrepancy after normalization. These continuity and stability statements are consequences of the raw formula; neither a Fisher metric nor an assumed covariance geometry enters. Theorem 3.2 now gives (6.3). □

6.2. **Noncommuting covariant quantum instruments.** Let $q \ge 2$. A physical state is a density operator on $\mathbb{C}^q$. Let ν be the unitarily invariant probability on rank-one projections P_ψ. It is a probability on complex projective space, not on vectors with an observable phase. Each control selects a channel

$$\mathcal{T}_a^\theta(\rho) = \eta_a^\theta \frac{I}{q}\operatorname{tr}\rho + (1-\eta_a^\theta)U_a^\theta \rho (U_a^\theta)^*, \qquad \eta_a^\theta \in [\eta_0, 1], \quad \eta_0 > 0, \tag{6.5}$$

where the fixed coefficients and unitaries may be unknown. The raw instrument is

$$L_a^\theta(d\psi)\rho = qP_\psi \mathcal{T}_a^\theta(\rho)P_\psi\,\nu(d\psi) = q\operatorname{tr}(P_\psi \mathcal{T}_a^\theta(\rho))P_\psi\,\nu(d\psi). \tag{6.6}$$

It is a completely positive instrument: composition with $\mathcal{T}_a^\theta$ is followed by the positive measurement map $\rho \mapsto qP_\psi\rho P_\psi$. Since $\int P_\psi\,d\nu = I/q$, its integrated trace is preserved. The report is P_ψ and the normalized output state is P_ψ.

After stopping, a protected covariant measurement on that output state reports P_ζ with density $q\operatorname{tr}(\rho P_\zeta)$ relative to ν. The numerical audit vector, regarded in the real Hilbert space of Hermitian matrices, is

$$Z = (q+1)P_\zeta - I. \tag{6.7}$$

The task is its Hilbert–Schmidt squared prediction loss. No further copy or quantum memory is available to the observer.

Theorem 6.2 (Quantum realization)**.** *The prepared instrument* (6.6) *and audit* (6.7) *verify Theorem 3.2(i)–(iii), with*

$$a_* = \eta_0, \qquad b_* = q, \qquad B = q^2 + q - 2, \qquad s = 2(q-1).$$

Preparation at I/q realizes the reference law ν in one call for every model. Uniformly over the allowed channel family,

$$\mathcal{R}^{\Theta}_{\mathrm{aut}}(N, M) - (q^2 + q - 2) \asymp \big(N(M-1)\big)^{-1/(q-1)}. \tag{6.8}$$

The instrument has continuously many, generally noncommuting, state-dependent effects. No shared eigenbasis or finite designed measurement ensemble is required.

Proof. Use the real space of Hermitian matrices, its positive semidefinite cone and trace as mass. Positivity, compactness of the density-operator base and finite-dimensional affine closure follow directly from (6.6). The likelihood obeys

$$\eta_0 \le q \operatorname{tr}(P_\psi \mathcal{T}^\theta_a(\rho)) \le q$$

for every density operator, hence for every incoming posterior mixture and adaptive history. The normalized map is the reported P_ψ, so it is report-recoverable. The action family changes probabilities, not the observer's access to the model.

We give the audit and geometry calculations. Unitary invariance, or the fourth moments of a normalized complex Gaussian vector, gives

$$\int P_\zeta \otimes P_\zeta \, \nu(d\zeta) = \frac{I + \mathsf{Swap}}{q(q+1)}. \tag{6.9}$$

To verify this formula directly, independent standard complex Gaussian coordinates have independent exponential squared moduli. Their normalized squared moduli have uniform simplex density, with second moments $2/(q(q+1))$ on the diagonal and $1/(q(q+1))$ off the diagonal; independent phases annihilate the remaining entries. Pairing (6.9) with ρ yields

$$\mathbb{E}[P_\zeta \mid \rho] = \frac{\rho + I}{q+1}, \qquad \mathbb{E}[Z \mid \rho] = \rho.$$

Moreover, Z has eigenvalues $q, -1, \dots, -1$, so $\|Z\|^2_{\mathrm{HS}} = q^2 + q - 1$. On each acquired pure state, $\|\rho\|^2_{\mathrm{HS}} = 1$, giving constant Bayes risk $B = q^2 + q - 2$. The audit separates all density operators, since its mean is ρ. It therefore supplies executable informationally complete tests, rather than an assumed state metric.

For a fixed ray P_ψ, put $V = |\langle \psi, \zeta \rangle|^2$. The same exponential-coordinate calculation gives

$$\mathbb{P}(V \ge u) = (1-u)^{q-1}, \quad 0 \le u \le 1.$$

Since $\|P_\psi - P_\zeta\|^2_{\mathrm{HS}} = 2(1-V)$, the actual reference measure has the exact ball mass

$$\nu\{P_\zeta : \|P_\psi - P_\zeta\|_{\mathrm{HS}} \le r\} = \left(\frac{r^2}{2}\right)^{q-1}, \quad 0 \le r \le \sqrt{2}. \tag{6.10}$$

This proves the required regularity on the acquired projective support. In particular, its dimension is $2(q-1)$, not the ambient real dimension q^2 of Hermitian coordinates.

Finally, trace-norm contraction of a channel and the elementary bound $|\operatorname{tr}(PA)| \le \|A\|_1$ give report-law continuity,

$$\|p^\theta_{a,\rho} - p^\theta_{a,\sigma}\|_{\mathrm{TV}} \le \frac{q}{2}\|\rho - \sigma\|_1.$$

The posterior for a common report is exactly the same projection. Thus the raw likelihood, actual mass, global cover, update stability and protected audit conditions have all been verified. Apply Theorem 3.2. □

Remark 6.3 (The role of physical preparation). The quantum output projection and the classical next state are states of the experiment, not extra registers of the observer. The observer retains only its current classical label. A call to the quantum instrument includes the displayed controlled channel and measurement; the number of such calls is τ. Gate counts, channel preparation, the protected audit call and the fixed apparatus dimension are additional physical costs. They are not equated with $\lceil \log_2 M \rceil$ classical bits. The rate is a statement for this declared instrument interface, not for arbitrary laboratory implementations with free resets.

7. Singular acquisition and changing task strata

The singular independent-preparation and nonconstant-Bayes-risk mechanisms are inherited from [7]. Here the regularity proof applies to all real mean budgets and the comparison survives correlated model families; the raw changing-rank instrument below supplies a separate realization of the weighted profile.

7.1. **A noninteger-dimensional raw realization.** Let μ_C be the centered uniform middle-thirds Cantor probability on $[-1/2, 1/2]$, and let $\nu = \mu_C^{\otimes d}$. This is an actual preparation law for (6.1); it is not Lebesgue measure restricted to a null set.

Proposition 7.1 (Singular controlled acquisition)**.** *For the classical raw experiment with $\nu = \mu_C^{\otimes d}$, Theorem 3.2 gives*

$$\mathcal{R}^{\Theta}_{\mathrm{aut}}(N, M) - \frac{d}{2} \asymp \big(N(M-1)\big)^{-2/(d \log 2/\log 3)}. \tag{7.1}$$

The conclusion holds for the full controlled correlated coefficient family, with all failed and accepted acquisitions charged.

Proof. A level-n Cantor cylinder has length 3^{-n} and mass 2^{-n}. If $3^{-(n+1)} < r \le 3^{-n}$, a ball centered on the Cantor set contains the level-$(n+1)$ cylinder containing its center, so its mass is at least $2^{-(n+1)}$. An interval of radius r meets at most a fixed number of level-n cylinders, since these intervals have length 3^{-n} and disjoint interiors separated by gaps at least 3^{-n}. Its mass is at most $C2^{-n}$. Thus $\mu_C(B(x,r)) \asymp r^{\log 2/\log 3}$ uniformly in x and r.

In d dimensions, a Euclidean ball contains the product of coordinate balls of radius $r/\sqrt{d}$ and is contained in the product of coordinate balls of radius r. Multiplying the one-dimensional estimates proves (3.11) with $s = d \log 2/\log 3$. The kernel, finite test closure, audit and model-uniform comparison were proved for any centered ν in Theorem 6.1; no density relative to ambient volume was used there. The general regular-support theorem therefore proves (7.1). □

A mixture $\nu = w\delta_0 + (1-w)\mu_C^{\otimes d}$ is also centered and is an actual reference preparation in the same experiment. The mass-constrained profile, not a single regular exponent, applies. In the independent reference member, $\mathcal{Q}_1^{\nu}(N) = 0$ once $N \ge 1/w$. The mean cost of selecting the atom is precisely $1/w$; its small physical probability is not erased by normalization. For the whole correlated family, the generic comparison gives a valid sufficient budget $1/(a_* w)$. In this particular centered kernel, the atom's next-call probability is exactly w at every physical state, so direct rejection gives the sharper budget $1/w$ as well.

7.2. **Nonconstant conditional Bayes risk.** Changing-rank quantum outputs naturally have different irreducible variances. Subtracting one common number would change the problem. There is an exact extension that retains them. Let $X = f_a(Y) \in K$, and suppose the protected squared audit has conditional Bayes risk $\beta(X)$, where $\beta \ge 0$ is known and continuous on compact K. Define

$$\mathcal{Q}^{\nu}_{m,\beta}(t) = \min_{C,\, 0 \le h \le t,\, \int h = 1} \int (\beta(x) + d_C(x)^2) h(x) \nu(dx). \tag{7.2}$$

Corollary 7.2 (Stratum-sensitive total risk)**.** *Under the report comparison hypotheses, with this nonconstant conditional Bayes risk, the checkpoint and autonomous* total *risks obey the bounds of Theorem 3.2(ii), with $\mathcal{Q}^{\nu}_{m}$ replaced by $\mathcal{Q}^{\nu}_{m,\beta}$ and with no subtraction of B. For independent reference reports,*

$$\mathcal{R}_{\mathrm{cp}}(N,m)=\mathcal{Q}^{\nu}_{m,\beta}(N),\qquad \mathcal{R}_{\mathrm{aut}}(N,M)=\mathcal{Q}^{\nu}_{M-1,\beta}(N). \tag{7.3}$$

The model-uniform upper factor remains b_*/a_* when $a_*N\geq 1$.

Proof. Replace the nearest-center cost d_C^2 in Lemmas 5.1 and 5.3 by the nonnegative continuous cost $\beta+d_C^2$. Compactness and threshold selection are unchanged. Conditional audit projection gives total risk $\mathbb{E}[\beta(X_\tau)+\|X_\tau-r_J\|^2]$. Lemma 5.2 proves the lower, and the same acceptance rule proves the upper and the independent-report equality. No comparison of different Bayes baselines has been made. □

7.3. **Raw quantum instruments with changing rank.** For $1\leq r\leq q$, let ν_r be Haar probability on the orbit

$$K_r=\{UP_rU^*/r : U\in \mathrm{U}(q)\},$$

where P_r is a fixed rank-r projection. The case $r=q$ is the single state I/q. Choose fixed weights $w_r\geq 0$, $\sum_r w_r=1$, with at least two positive weights. A raw call reports the marked state (r,σ) and prepares σ through

$$L^{\theta}_a(r,d\sigma)\rho=w_r q\,\mathrm{tr}(\sigma\mathcal{T}^{\theta}_a(\rho))\sigma\,\nu_r(d\sigma). \tag{7.4}$$

This is a positive, indeed completely positive, measure-and-prepare instrument. Each effect is positive; the prepared state is positive; and $\int\sigma\,d\nu_r=I/q$ proves integrated trace normalization. The same protected covariant audit (6.7) has conditional mean σ and

$$\beta(\sigma)=q^2+q-1-\mathrm{tr}\,\sigma^2=q^2+q-1-\frac{1}{r}. \tag{7.5}$$

The finitely many rank orbits are disjoint compact sets, and this function is continuous on their union. No observer register containing r is required after the terminal label has been selected; r is part of the current report only.

Proposition 7.3 (Changing-rank acquired profile)**.** *The raw instrument* (7.4) *satisfies the general finite-affine theorem and Corollary 7.2 with actual reference law $\nu=\sum_r w_r\nu_r$, $a_*=\eta_0$ and $b_*=q$. In the independent member $\eta^{\theta}_a=1$ for all actions, its exact autonomous risk is*

$$\mathcal{R}_{\mathrm{aut}}(N,M)=\min_{C,\,h}\sum_r w_r\int_{K_r}\left(q^2+q-1-\frac{1}{r}+d_C(\sigma)^2\right)h(\sigma)\nu_r(d\sigma), \tag{7.6}$$

where $0\leq h\leq N$ and $\sum_r w_r\int h\,d\nu_r=1$, and $|C|\leq M-1$. This covers mixtures of continuous rank strata and a genuine full-rank atom without conditioning away any weight.

Proof. Use the positive semidefinite cone and trace. Formula (7.4) is linear in the incoming density matrix and has normalized output equal to the report. The density relative to ν lies between η_0 and q, because $\mathrm{tr}\,\sigma=1$ and $\mathcal{T}^{\theta}_a(\rho)\geq\eta_0 I/q$. Preparation at I/q gives exactly ν. The audit calculations establishing its affine moments and separation are those already proved in Theorem 6.2, valid for every density operator. Report-kernel continuity follows by the same trace-norm bound, and same-report normalization is independent of the incoming state. Thus all hypotheses are verified from the raw maps. Corollary 7.2 proves (7.6). □

The formula optimizes the genuine task, including its different rank-dependent Bayes risks. The zero-distortion atom criterion in Proposition 5.5 concerns the distortion component; it is not a claim that selecting the full-rank atom minimizes (7.6). This distinction is material when the stopping decision changes the irreducible audit risk.

8. Same-task error composition and paid implementation

First-discrepancy coupling, state-product simulation, and attained hidden-offset and erased-bit constructions are inherited mechanisms [7, 6]. We state their precise form for the present mean-budget architecture, and combine them with its correlated acquired-geometry law.

8.1. **Stopped-law perturbations.** A numerical or calibration comparison must include the stopping decision and the protected audit. Approximation of the report marginal alone does not compare the scored task.

Proposition 8.1 (Acquired-occupation error composition)**.** *Consider two causal implementations on the same visible interface. Suppose their next conditional interface kernels, on a common pre-acquisition history h, admit a coupling with mismatch probability at most $\kappa(h)$; the protected final audit coupling has mismatch at most $\kappa_f(h)$. Use the same observer and fresh randomization in both. If the ideal implementation has finite mean stopping time, then the total variation distance of their complete stopped, scored laws is at most*

$$\min\left\{1,\ \mathbb{E}\sum_{t=1}^{\tau}\kappa(H_{t-1})+\mathbb{E}\kappa_f(H_\tau)\right\}. \tag{8.1}$$

A possible nonterminating approximate run is recorded at an extra failure outcome. If $0\le \ell\le L$, the corresponding risk difference is at most L times (8.1)*. In particular, uniform bounds $\kappa\le\theta$, $\kappa_f\le\theta_f$ and $\mathbb{E}\tau\le N$ give $L(N\theta+\theta_f)$, truncated at L.*

Proof. Couple the conditional kernels sequentially while the histories agree, and use identical routing coins. The event that the first disagreement occurs at acquisition t has probability at most the ideal expectation of $\mathbf{1}_{\{\tau\ge t\}}\kappa(H_{t-1})$. After the ideal stop, couple the protected audit in the same way. The union bound and monotone convergence give (8.1); on its complement the complete finite stopped records and audit outcomes agree. Coupling bounds total variation with convention $\|P-Q\|_{\mathrm{TV}}=\sup_A|P(A)-Q(A)|$. Integrating a loss in $[0,L]$ gives the risk bound. This argument concerns the conditional kernels of the full compared interface, including the hidden-state effect needed for later reports or its equivalent visible conditional law. A bound on a single current-report marginal is not substituted for that hypothesis. □

Remark 8.2 (Mean cost is a separate obligation)*.* Total variation does not control expected acquisition. Replacing a one-call stop by a branch of probability ε lasting ε^{-2} calls changes the complete-law probability of that branch by at most ε, while adding order ε^{-1} to the mean. Thus (8.1) is not a budget certificate for the approximate machine. For the rejection construction one may plan an ideal conditional stop floor $(1+\zeta)/N$ and require the implemented acceptance probability to differ by at most ζ/N at every surviving history. The implemented floor is then $1/N$, which proves its own mean bound and geometric tail. Appendix B gives a geometric alternative with a deterministic acceptance margin.

If a fixed acceptance rule is retained but each final code vector is perturbed by at most δ_{out}, Minkowski's inequality gives

$$\mathbb{E}\|X_\tau-\widehat{r}_J\|^2\le\left(\sqrt{\mathbb{E}\|X_\tau-r_J\|^2}+\delta_{\mathrm{out}}\right)^2. \tag{8.2}$$

Only when $r_J=\mathbb{E}[X_\tau\mid J]$ under the same actual stopped law does orthogonality improve this to addition of δ_{out}^2. Centers optimized for a reference law need not be calibrated centroids for every unknown correlated model. The stronger additive formula is therefore not asserted for them.

8.2. **Resource-certified causal simulation.**

Proposition 8.3 (Complete-task causal transfer)**.** *A parameter-independent causal simulator from a source experiment to a target experiment has at most R internal states and a causal action lift. It transforms reports, stopping, and the protected task readout compatibly. Suppose that, uniformly over models and all target policies of interest, its complete stopped scored law has total variation error at most ε, and its source acquisition certificate is*

$$\mathbb{E}[\textit{source calls}] \le \kappa \mathbb{E}[\textit{target calls}] + c.$$

Then a target observer with M states and mean acquisition at most N has a source implementation with at most RM states and mean acquisition at most $\kappa N + c$, satisfying

$$\mathcal{R}_{\mathrm{source}}(\kappa N + c, RM) \le \mathcal{R}_{\mathrm{target}}(N, M) + L\varepsilon. \tag{8.3}$$

Two such simulators compose with error at most $\varepsilon_1 + \varepsilon_2$, state multiplier R_1R_2, and acquisition bound

$$N \longmapsto \kappa_1\kappa_2 N + \kappa_1 c_2 + c_1. \tag{8.4}$$

Program descriptions, temporary workspaces and physical execution times remain separately charged.

Proof. Run the simulator and target observer in their product state. Only past reports and fresh coins enter the product transition. The action lift ensures that every requested target action has a legal source realization; compatibility of the stop and audit prevents information from arriving after the target decision but before the source decision. Apply the declared mean-cost certificate to the executed target policy and the total variation loss bound to the complete task. Infimizing over target observers proves (8.3). Sequential composition gives (8.4) by substitution. Kernel contraction of total variation and the triangle inequality give the additive error. The product of internal state sets gives the state count. Reverse lower bounds require a reverse simulator or an independent lower argument; they do not follow from this one-sided construction. □

8.3. **An attained joint resource–risk law.** It is useful to distinguish an allowable error upper bound from an experiment that actually has that error. We give the latter in the same task. Start with a regular-support report-recoverable experiment from Theorem 3.2. A stopped core state is frozen. The protocol then requires exactly one bit acquisition: for an unknown model bit $b \in \{-1, 1\}$ it reports b with probability $1 - e$ and an erasure symbol with probability e. The bit affects no preceding core report. An unknown calibration parameter $c \in \{-\delta, \delta\}$ affects no observer report at all. The final protected audit returns the core audit, b, and c, and the loss is the sum of their squared prediction losses. The observer must commit all three predictions before this audit. It is forbidden to restart the core or repeat the bit acquisition.

Lemma 8.4 (Attained erasure deficiency)**.** *For this bit interface, relative to the perfect bit experiment, the minimum parameter-independent causal simulation error is exactly $\varepsilon = e/2$. The protected truth readout is fixed and cannot be altered by the simulator. The minimax bit prediction risk is e, and the independent hidden-calibration prediction risk is δ^2.*

Proof. On a nonerased report, output the reported bit. On erasure, a fair guess gives error $e/2$ for each model bit, so it achieves the stated total variation defect against the perfect report and unchanged protected truth. Conversely, under the uniform prior on $b = \pm 1$, no pre-audit information on an erased branch depends on b. Every simulator then guesses wrongly with probability at least $1/2$ on that branch. Its largest model error is at least its prior average, $e/2$. This proves exact deficiency, without access to the final audit. For squared bit prediction, the conditional Bayes estimator on erasure is zero and its loss is one; the minimax risk is e, attained by zero on erasure and the reported sign otherwise. Since the calibration sign is absent from every report, averaging its two signs gives $\mathbb{E}(r - c)^2 = \mathbb{E}r^2 + \delta^2$, minimized at zero. □

Theorem 8.5 (Attained acquisition–memory–error composition)**.** *Let $0 \le \delta \le 1$, $0 \le \varepsilon \le 1/2$, $n \ge 1$ and $M \ge 5$. Give the core a mean acquisition budget n and use the extended protocol with erasure probability 2ε. Its model-uniform optimum satisfies*

$$\mathcal{R}_{\mathrm{ext}}(n, M, \delta, \varepsilon) - B \asymp (nM)^{-2/s} + \delta^2 + \varepsilon. \tag{8.5}$$

The upper is implemented by at most $4m + 1 \le M$ states, with $m = \lfloor (M-1)/4 \rfloor$. Under the convention charging one preparation, each core call, one bit call and one protected audit call, the mean total physical-call budget is $\mathsf{N} = n + 3$. Thus the geometric term in that convention is $((\mathsf{N}-3)M)^{-2/s}$, not $(\mathsf{N}M)^{-2/s}$ with the extra calls silently omitted.

Proof. For the upper, use the one-live-state core construction with m centers. Upon acceptance, retain its center index in one of m bit-waiting states. These states perform the single bit call. Its three possible reports lead to $3m$ terminal states, carrying the fixed core center and the bit prediction in $\{-1, 0, 1\}$. The calibration prediction is zero. The total is $1 + m + 3m = 4m + 1$. No clock, free retained bit or exact posterior is present. The core risk bound and Lemma 8.4 give

$$\mathcal{R}_{\mathrm{ext}} - B \le C(nm)^{-2/s} + \delta^2 + 2\varepsilon.$$

Since $m \ge M/8$ for $M \ge 5$, this is the upper in (8.5).

For the converse, any legal extended observer commits to a core stop before receiving the bit. Its actual stopped core law is bounded by $b_* n\nu$. The first component of its final decoder has at most M possible values, so its core squared distortion is at least $\mathcal{Q}_M^{\nu}(b_* n) \ge c(nM)^{-2/s}$. This remains true even though the later bit report can select among those values. Average over the two bit signs and two calibration signs, keeping any one core model fixed. These signs do not change the preceding core law. Conditional on an erased bit there is still no information about its sign; there is never pre-audit information about the calibration sign. The three squared losses add, so their averaged lower bound is $c(nM)^{-2/s} + 2\varepsilon + \delta^2$. The worst model risk is at least this average. The physical-call count is a direct count of the specified protocol. There is no conversion from total variation to an expected-cost bound in this proof. □

Theorem 8.5 supplies attained lower and upper error terms for a fixed enlarged task. Proposition 8.3 is the general transfer statement. They are logically different results: an arbitrary permitted calibration tolerance or simulator error need not be attained by a given experiment and hence need not cause that much risk.

Appendix A. Executable tests and measurable predictive quotients

We give the elementary finite-closure construction underlying the representation in Section 2. Fix one prepared model. Let $\mathcal{V}$ be the linear space generated by bounded executable future-test functions on the physical state, including the constant function, the audit moments, and all legal one-step report pullbacks. A test is a specified causal continuation with a bounded final report, not an arbitrarily attached coordinate. Suppose $\mathcal{V}$ is finite-dimensional and admits the positive instrument representation (2.1). Choose a basis $1, f_1, \dots, f_d$ and evaluate it on prepared posterior measures.

Proposition A.1 (Realization and minimality)**.** *Two prepared posterior measures give the same value to every test in $\mathcal{V}$ if and only if their basis evaluation vectors agree. Finite future report-and-action laws and the protected audit descend to this evaluation. On positive evidence, its update is* (2.2)*. The history-to-evaluation map is Borel and gives a standard Borel realization in the compact base D. Any Borel statistic of a standard Borel history that determines all these future tests factors this evaluation through a Borel map.*

This does not assert that the image of every Borel history map is a Borel subset of D. When only an analytic image is established, D is the declared Borel realization and probability laws are

pushforwards into D; unused points are not declared acquired. In the two raw realizations the post-report evaluation images are compact and therefore Borel.

Proof. Equality on a finite basis is equivalent to equality on its span. Pullback closure means that the probability of a report event and the unnormalized numerator of every subsequent test are linear evaluations of the incoming vector. Dividing by its positive report likelihood gives (2.2); zero likelihood has zero positive numerator by strict positivity of e on the cone. Iterating gives every finite continuation law. Bounded monotone limits of such tests remain affine when they belong to the declared closure: evaluation on a fixed finite spanning family of prepared states determines the coefficients, so pointwise bounded convergence determines their limit. Alternatively, one may retain just the explicitly declared finite continuations; no closure under an unproved infinite-horizon operation is needed for the main theorem.

The posterior evaluation is Borel by standard Borel conditional kernels, or directly by finite matrix products and normalization on the finite-report history spaces, pieced together over their countable union. If another statistic $r(h)$ determines all tests, each evaluation coordinate is constant on its fibers. For completeness, the Borel factor statement follows by separating the two disjoint analytic images under r of the Borel sets where a coordinate is, respectively, at most or greater than a fixed rational number. This shows that each coordinate is measurable with respect to $\sigma(r)$. The measurable factorization lemma for a real-valued function, applied coordinatewise, supplies a Borel factor; the resulting factor is defined on the entire standard Borel codomain of r. Its inverse image of the closed set D is Borel and contains every attained statistic; replacing its values on the complement of that Borel set by a fixed point of D makes it D-valued. No modification on a potentially non-Borel image complement is used. Thus the evaluation is the minimal information required by these tests, in this precise factorization sense.

The image caveat follows because Borel images of standard Borel spaces are in general only analytic. The general construction does not require the image to be the state space of the machine. In Theorem 6.1, the map $x \mapsto \phi(x)$ is continuous and injective on compact K, and the audit probabilities separate the full convex base. In Theorem 6.2, the density matrices form a compact set and the informationally complete audit separates them. Their acquired rank orbits are also compact. These give the stronger image assertion for the realizations. □

For a family of unknown models, this construction is made within each model's finite predictive representation. A posterior on an unknown continuous model parameter can require additional, even infinite-dimensional, coordinates. Theorem 3.2(i) does not pretend otherwise: it couples the model-indexed certificates by a single observer kernel. It never gives that kernel access to a model-specific posterior. The geometric upper is stronger in a different respect: the current response $f_a(y)$ and the acceptance rule are common to the whole family.

Example A.2 (Separate feasibility does not give a common program)**.** Prepare a fair bit U. In model 0 the report is $Y = U$; in model 1 it is $Y = 1 - U$. The protected audit is U. Both models have the same fair report law, and each separately has a perfect one-call decoder. No common decoder can be perfect: under the uniform prior on the two model labels, U and Y are independent fair bits. For squared 0/1 prediction the common minimax risk is $1/4$, attained by reporting $1/2$. Maximizing the routing contribution separately within each model would incorrectly certify zero. This is why model contributions in (3.2) are added before its maximum, and why terminal readouts remain common.

Appendix B. Finite descriptions and numerical resource certificates

The exact optimization in Theorem 3.2(i) is over Borel kernels. It is a causal realization theorem in the stochastic-kernel model, not a theorem that every such kernel has a finite Turing description. We now give an independent finite-description upper for the sharp regular-support class. Its extra input is a certified numerical report and a computable geometric net, not a stored exact posterior.

Proposition B.1 (Finite-bit regular-support implementation)**.** *Assume Definition 3.1 and* (3.11)*. Suppose the following data are supplied by the raw implementation: at accuracy 2^{-p}, the current task response in $\mathbb{R}^k$ can be read with deterministic Euclidean error at most $C_y 2^{-p}$ using workspace $W_y(p)$ and time $T_y(p)$; finite rational global covers and separated center lists at their asserted scales can be constructed; and the fixed action a_0 has a finite description of length D_0.*

Then a finite program attains the upper order in (3.13)*, using one live label and $m = M - 1$ terminal labels, with*

$$p \le C + \left\lceil \frac{1}{s} \log_2(Nm) \right\rceil, \qquad D_{\mathrm{prog}} \le D_0 + Ckm(p + \log(m+1)). \tag{B.1}$$

One valid temporary workspace bound is

$$W_{\mathrm{temp}} \le W_y(p) + Ck(p + \log(k+1)) + C\log(m+1), \tag{B.2}$$

and each live transition uses at most $T_y(p) + Cm\,T_{\mathrm{dist}}(k,p)$ arithmetic time. The constants include the fixed geometry and certified arithmetic implementation. The mean number of transitions is at most N; there is no deterministic upper bound on the number of transitions. The program and report computation do not access the unknown model.

Proof. For sufficiently large N take m separated centers as in Lemma 5.4. Set $r = C_2(Nm)^{-1/s}$, where C_2 is a fixed large constant. Choose the large-N threshold so that $3r$ is less than their separation. Round the centers and read the current response accurately enough that their combined distance error, including certified arithmetic, is at most $r/4$. Accept when the computed distance to some rounded center is at most r. The true union of radius-$r/2$ balls is then contained in the accepting set, and every accepted point is within $2r$ of its selected rounded decoder. The lower acquired mass bound implies, at every surviving history,

$$\mathbb{P}(\text{accept} \mid h) \ge a_* m c_0 (r/2)^s \ge 1/N$$

by the choice of C_2. Thus the *implemented* stopping time has mean at most N, independently of any total variation approximation. Its squared distortion is at most $4r^2$.

For N below the fixed threshold, use a certified global cover with m centers, stop after the first report, and select a center of approximately smallest distance. The cover and numerical errors give $Cm^{-2/s} \le C'(Nm)^{-2/s}$ in this range. Both cases use $p = O(1 + s^{-1}\log(Nm))$. Storing m rounded k-vectors, a fixed threshold and their indices gives (B.1), allowing harmless additional index-address bits. Computing distances by scanning the read-only list requires one current response, one running best distance, one index, and arithmetic scratch space, which gives (B.2). This workspace is erased before a rejected call returns to the single live label. The terminal center index is the only retained outcome. Time is counted for every attempted call, including failures. □

Certified nets in the two realizations. For the box, rational lattice covers and separated sublattices provide the required lists. For the product Cantor set, depth-h cylinders have rational endpoints, known mass 2^{-dh} and known separation at their natural scale. Taking enough endpoints supplies separated lists, and complete cylinder families supply covers. The relation $3^{-h} \asymp m^{-1/s}$, with 2^{dh} within a fixed factor of m, gives the asserted cardinality bounds. No membership oracle for an arbitrary null set is assumed.

For the quantum support, cover projective space by the q charts in which $|\psi_j|$ is largest. After fixing that coordinate's phase, write the ray as $(1,z)/\|(1,z)\|$, with all real and imaginary components of z in $[-1,1]$. The corresponding projection

$$P(z) = \frac{(1,z)(1,z)^*}{1 + \|z\|^2}$$

is Lipschitz on each compact chart with a constant depending only on q. Rational grids of mesh h therefore give computable covers of cardinality $O_q(h^{-2(q-1)})$ and rational matrix coordinates. A

greedy separated subset of a sufficiently fine such net supplies the packing lists: maximality gives a cover at a fixed multiple of the separation scale, and the exact mass formula (6.10) bounds its cardinality above and below. Choosing the scale with a fixed safety factor produces at least m separated points, or at most m covering points, as needed. These are finite, verifiable constructions, not a claim of a fast optimal projective quantizer.

The literal description length in (B.1) grows with precision even though the observer has only M persistent labels. For the quantum instrument, its fixed physical dimension, channel invocation, outcome digitization and audit measurement are additional native costs. For the classical instrument, the vector-report acquisition and the scalar operations implementing it are likewise distinct. A bounded-state theorem alone cannot identify either of these total bit or physical cost frontiers.

What is not supplied by a finite check. The support criterion (3.2) quantifies over real dual vectors and, for an arbitrary model family, all finite model subsets. The proof establishes the continuum statement by compactness and separation. Exact finite examples check special instances, signs and constants; they do not replace that proof. Nor does the geometric program calculate the optimal centers in (3.4): it implements the proven matching order under explicit numerical certificates. Optimal program bits, workspace, planning time and simulator-state lower bounds remain separate mathematical questions.

Appendix C. Connection with informative Bayesian compression

The sharp free-stopping profile is not a replacement for the more general informative-report identities. We record the latter to specify exactly what is inherited and what extra hypothesis the profile uses. The identity below is the normalized-pooling result of the separately retained Bayesian-curvature article [6]; its short proof is included for independence of notation.

Let $U(y,x)$ and $\ell(y,x)$ be affine in a finite predictive vector x, with $\ell \geq 0$, normalized report law and bounded posterior $F = U/\ell$ on positive evidence. Let X be the exact incoming vector under an actually prepared law, S its retained label, and $V = \mathbb{E}[X \mid S]$. Write X' for the exact next vector, Y for the current report and J for any randomized outgoing label using only (S, Y) and fresh coins. Define

$$\Psi(x) = \int \frac{\|U(y,x)\|^2}{\ell(y,x)} \lambda(dy), \quad W = F(Y,V), \quad c_{sj} = \mathbb{E}[W \mid S = s, J = j], \quad v_j = \mathbb{E}[W \mid J = j],$$

with zero contribution from zero-evidence reports.

Proposition C.1 (Inherited three-defect identity)**.** *Under the displayed raw affine and measurability assumptions,*

$$\begin{aligned}\mathbb{E}\|X' - \mathbb{E}[X' \mid J]\|^2 = \mathbb{E}\Psi(X) - \mathbb{E}\Psi(V) \\ + \mathbb{E}\|W - c_{SJ}\|^2 + \mathbb{E}\|c_{SJ} - v_J\|^2.\end{aligned} \tag{C.1}$$

All three right-hand terms are nonnegative. The first equals $\mathbb{E}\|X' - \mathbb{E}[X' \mid S, Y]\|^2$.

Proof. Affine unnormalized integration gives the report density $\ell(y, V)$ conditional on S and the corresponding numerator $U(y, V)$. Thus $\mathbb{E}[X' \mid S, Y] = W$, with the necessary likelihood reweighting already included. Fresh routing randomization gives $\mathbb{E}[X' \mid S, Y, J] = W$. Consequently $\mathbb{E}[X' \mid J] = v_J$. Two nested conditional squared-norm decompositions yield

$$\mathbb{E}\|W\|^2 = \mathbb{E}\|W - c_{SJ}\|^2 + \mathbb{E}\|c_{SJ} - v_J\|^2 + \mathbb{E}\|v_J\|^2.$$

Integrating the raw report kernels gives $\mathbb{E}\|X'\|^2 = \mathbb{E}\Psi(X)$ and $\mathbb{E}\|W\|^2 = \mathbb{E}\Psi(V)$. Subtract the last term of the preceding display to obtain (C.1). Conditional projection proves the claimed form and sign of the first difference. No differentiability is needed. □

Rowwise optimal quantization does not imply equality of the pooled centroids c_{sj}, and one-step equality does not imply that compatible rows can be chosen throughout an arbitrary future experiment. For a general filter, the occupation criterion of Theorem 3.2(i) enforces that a single whole-run observer exists. Recoverability is the additional raw condition that makes the current task vector $W = f_a(Y)$ known from the current report and permits the sharp free-stopping construction. The retained informative Gaussian and noncommuting weak-effect filtering proofs use a different mechanism, namely actual mass, global covers and summable continuation sensitivity. They remain independent positive results; their fixed-task or prescribed-stop conclusions are not automatically statements about optimization under a free mean budget.

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